La Consolacion University Philippines
Valenzuela St., Capitol View Park Subdivision, Bulihan, City of Malolos, Bulacan

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SCHOOL YEAR 2025-2026
COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING

I. COURSE CODE : STATIE2

II. COURSE TITLE : Statistical Analysis for Industrial Engineer II

III. COURSE DESCRIPTION :

This course (ES) is designed for undergraduate engineering students with emphasis on problem solving related to societal issues that engineers and scientists are called upon to solve. It introduces different methods of data collection and the suitability of using a particular method for a given situation.

The relationship of probability to statistics is also discussed, providing students with the tools they need to understand how plays a role in statistical analysis. Probability distributions of random variables and their uses are also considered, along with a discussion of linear functions of random variables within the context of their application to data analysis and inference.

The course also includes estimation techniques for unknown parameters; and hypothesis testing used in making inferences from sample to population; inference for regression parameters and build models for estimating means and predicting future values of key variables under study. Finally, statistically based experimental design techniques and analysis of outcomes of experiments are discussed with the aid of statistical software.

Aligned with the Marian Augustinian values, this course integrates ethical considerations and encourages students to approach technology with a balanced view of its potential to both advance and challenge societal norms. The program aims to foster holistic development, nurturing students not only as technical experts but as Christo-centric Catholic professionals with a commitment to ethical innovation and responsible stewardship of technology.

IV. PRE-REQUISITE : STATIE1

V. CREDIT UNITS : 3 units

VI. COURSE LEARNING OUTCOMES:

Vision	Mission
Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development of Catholic Augustinian Marian graduates rooted in Gospel values.	To achieve the vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them: ● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;* ● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;* ● *live the Gospel values in the varied aspects of their personal and professional lives;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* ● *develop individuals who are resilient, inclusive and responsive to be socio-economic realities of the Philippines and the complexities of the global environment.*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
Creative Critical Thinker	PO1. Ability to apply knowledge of mathematics and science to solve complex engineering problems; PO2. Ability to design and conduct experiments, as well as to analyze and interpret data; PO3. Ability to design a system, component, or process to meet desired	CLO1: Recognize one's own professional, social, and ethical responsibilities and gain the initiative to apply the different laws, methods and theorems of engineering data analysis to provide solutions particularly to the complex problems being faced in the field of engineering [EGA1] [PO1] [PO2]

Competent Catholic Augustinian Marian Professional Socially Responsible Steward Transformative Life Long Learners	needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability, in accordance with standards. PO4. Ability to function on multidisciplinary teams PO5. Ability to identify; formulate, and solve complex engineering problems; PO6. Understanding of professional and ethical responsibility. PO7. Ability to communicate effectively PO8. Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context. PO9. Recognition of the need for, and an ability to engage in life-long learning; PO10. Knowledge of contemporary issues; PO11. Ability to use techniques, skills and modern engineering tools necessary for engineering practice; and PO12. Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in PO13: Ability to design, develop, implement and improve integrated system	CLO2. Design experiments involving several factors (EGA 1 [EGA2] [PO6, PO7, PO12] CLO3: Engage in life-long learning and understanding and apply the principles of engineering data analysis in the field of engineering for the country's growth and development [EGA3] [PO4, PO8] CLO4: Work effectively and efficiently with the use of engineering data analysis principles in the various fields of engineering. [EGA4] [PO10, PO11, PO13]

	that include people materials, implementation, equipment and energy.	

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	COURSE CONTENTS	SUGGESTED LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUE/S TO BE INTEGRATED
Week 1	ILO1 Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	**I. COURSE ORIENTATION** • Introduction to the Course Contents • Course Requirements and Activities • Course intended learning outcomes • Grading System and Policies on Absences, Tardiness and Examinations	• Setting and sharing of expectations from the course • Discussion of the Vision, Mission and Goals of LCUP and of CITE • Presentation of course • intended learning outcomes and contents, course requirements and activities • Review of policies on absences, tardiness, examinations	• Statement of Expectations • Review of Course Contents, Objectives, Activities and Assessment Tools • Review of Policies on Absences, Tardiness, Examinations, Grading System	• Understanding • Open mindedness • Respect • Obedience • Interiority • Courage

			, grading system		
Week 2-3	ILO1: Apply statistical methods in the analysis of data [CLO1] ILO2. Design experiments involving several factors [CLO2] LO3: Work effectively and efficiently with the use of engineering data analysis principles in the various fields of engineering. [CLO3]	**I. Introduction to Statistical Test** 1.1 Null and Alternative Hypothesis 1.2 One-Tailed and Two-Tailed Hypothesis 1.3 P-Value in Hypothesis Testing 1.4 Types of Errors in Hypothesis Testing 1.5 Concluding a Statistical Test 1.6 Basic Components of a Statistical Test	• Lecture and class discussion using power point presentation • Discovery learning	• Recitation • Quiz • Seatwork/Assignment • Project Proposal	• **Understanding and patience in learning** • **Creative and critical thinking** • **Analytical thinking** • **Good and active participation and involvement** • **Appreciation** • **Effective and efficient management of resources**
	ILO1: Recognize one's own professional, social, and ethical responsibilities and gain the initiative to apply the different laws, methods and theorems of engineering data analysis to provide solutions particularly to the complex problems being faced in the field of engineering.[CLO2, CLO3] ILO2: Engage in life-long learning and	**II. Testing the Mean** 2.1 Test on the Mean of a Normal Distribution, Variance Known 2.2 Test on the Mean of a Normal Distribution, Variance Unknown 2.3 Test on the Mean Using Critical Regions (Traditional Method)	• Lecture and class discussion using power point presentation • Discovery learning	• Recitation • Quiz • Seatwork/ Assignment	• **Understanding and patience in learning** • **Creative and critical thinking** • **Analytical thinking** • **Good and active participation and involvement** • **Appreciation** • **Effective and efficient management of resources**

Week 4-5	understanding and apply the principles of engineering data analysis in the field of engineering for the country's growth and development. [CLO4] ILO6. Demonstrate the Filipino Catholic Augustinian Spirituality through value webbing. [CLO3]	2.4 Critical Region Method 2.5 Concluding Test Using Critical Region Method 2.6 Test Involving Paired Differences 2.7 Testing Paired Differences Using Student's t-Distribution			• **Problem solving skills**
Week 6		• **PRELIM EXAMINATION**		• **PRELIM EXAMINATION**	
Week 7-9	.ILO 1: Explain the concept of correlation and interpret scatter diagrams to determine the direction and strength of relationships between variables. *[CLO2], [CLO3]* **ILO 2:** Compute and interpret the correlation coefficient (r) to measure the degree of linear association between two variables. *[CLO2], [CLO3]* **ILO 3:** Test the significance of the correlation coefficient and draw inferences based on	**III. Correlation** 3.1 Scatter Diagram and Linear Correlation 3.2 Correlation Coefficient (r) 3.3 Formulation of Correlation Coefficient (r) 3.4 Computing Correlation Coefficient (r) 3.5 Testing Correlation Coefficient (r) 3.6 Inference for Correlation	• Lecture and class discussion using power point presentation • Discovery learning • Case Study Example	• Recitation • Quiz • Seatwork/ Assignment	• **Creative and critical thinking** • **Analytical thinking** • **Good and active participation and involvement** • **Appreciation** • **Effective and efficient management of resources** • **Problem solving skills**

	statistical evidence. *[CLO4]*				
Week 10-11	**ILO 1:** Explain the concept and purpose of linear regression in analyzing relationships between dependent and independent variables. *[CLO1]* **ILO 2:** Apply the least-squares criterion to derive the equation of a regression line. *[CLO2]* **ILO 3:** Evaluate the goodness of fit using the coefficient of determination (r^2) and standard error of estimate. *[CLO3]* **ILO 4:** Perform inference for regression to assess the significance of model parameters and validate the regression model statistically. *[CLO4]*	IV. **Linear Regression** 4.1 Linear Regression 4.2 Least-Squares Criterion 4.3 Finding the Equation of a Least-Squares Line 3.8 Using Least-Squares Line for Prediction: Interpolation 4.4 Using Least-Squares Line for Prediction: Extrapolation 4.5 Coefficient of Determination (r^2) 4.6 Inference for Regression 4.7 Standard Error of Estimate	• Lecture and class discussion using power point presentation • Discovery learning • Problem solving	• Recitation • Quiz • Seatwork/ Assignment	• **Creative and critical thinking** • **Analytical thinking** • **Good and active participation and involvement** • **Appreciation** • **Effective and efficient management of resources** **Problem solving skills**
Week 12	**MIDTERM EXAMINATIONS**			**MIDTERM RESULT**	
Week 13-15	. **ILO 1:** Apply the least-squares criterion to develop multi-linear regression models and	**V. Multi-linear Regression**	• Lecture and class discussion using power	• Recitation • Quiz • Seatwork/ Assignment	• **Understanding and patience in learning** • **Creative and critical thinking**

	compute regression coefficients using statistical software. [CLO2, CLO3] **ILO 2:** Evaluate the statistical significance and reliability of regression coefficients through hypothesis testing and confidence intervals. *[CLO3]*	4.1 Basic Terminology and Notation 4.2 Theory for Least-Squares Criterion 4.3 Regression Models and Computers 4.4 Testing Coefficient for Significance 4.5 Confidence Intervals for Coefficients	point presentation • Discovery learning • Problem solving		• **Analytical thinking** • **Good and active participation and involvement** • **Appreciation** • **Effective and efficient management of resources** **Problem solving skills**
Week 16-17	**ILO1:** Perform and interpret **Analysis of Variance (ANOVA)** for single-factor and two-factor experiments to determine the significance of treatment effects. *[CLO3, CLO4]* **ILO 2:** Apply randomized experimental designs to minimize bias and enhance the reliability of experimental results. *[CLO3, CLO4]*	**VI.Design of Experiment** 5.1 Analysis of Variance Single Factor 5.2 Analysis of Variance Two-way Factor 5.3 Randomized	• Lecture and class discussion using power point presentation • Discovery learning • Problem solving		• **Understanding and patience in learning** • **Creative and critical thinking** • **Analytical thinking** • **Good and active participation and involvement** • **Appreciation** • **Effective and efficient management of resources** • **Problem solving skills**
Week 18	**FINAL EXAMINATIONS**			**FINALS RESULT**	

VIII. COURSE REQUIREMENTS :

1. Term examinations
2. Quizzes
3. Seatwork, assignment and/or project
4. Class attendance
5. Class participation
6. Project

IX. REFERENCES:

BOOKS

Agresti, A., Franklin, C., & Klingenberg, B. (2018). *Statistics The Art And Science Of Learning From Data* (4th Ed.). Harlow, En: Pearson Education Limited.

Devore, J. L. (2016). *Probability And Statistics For Engineering And The Sciences* (9th Ed.). Boston, Ma: Cengage Learning.

Gould, R., Ryan, C., & Wong, C. (2017). *Essential Statistics* (2nd Ed.). Harlow, En: Pearson Education Limited.

Larson, R., & Farber, B. (2015). *Elementary Statistics Picturing The World* (6th Ed.). Boston, Ma: Pearson Education, Inc.

Montgomery, D. C., & Runger, G. C. (2014). *Applied Statistics And Probability For Engineers* (6th Ed.). Hoboken, Nj: John Wiley & Sons, Inc.

Quirk, T. J. (2016). *Excel 2016 For Engineering Statistics A Guide To Solving Practical Problem*. Sw: Springer International Publishing.

Spiegel, M. R., & Stephens, L. J. (2018). *Statistics* (6th Ed.). New York, Ny: Mcgraw-Hill Education.

Sullivan Iii, M. (2018). Statistics Informed Decisions Using Data (5th Ed.). Harlow, En: Pearson Education Limited.

Triola, M. (2018). *Elementary Statistics* (13th Ed.). Boston, Ma: Pearson Education, Inc.

Weiss, N. A. (2017). *Introductory Statistics* (10th Ed.). Harlow, En: Pearson Education Limited.

X. GRADING SYSTEM :

1. Class Standing 60%
   - Quiz
   - Recitation
   - Attendance
2. Examination 40 %
3. Term Grade CS (60%) + TE (40%)
4. Final Rating Prelim Grade + Midterm Grade + Final Grade

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5. **Passing Grade: 50%**

Prepared by:	Checked and Reviewed by:	Approved by:
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Revision Date:	August 23, 2025
Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	